WEB PROGRAMMING LAB-9 LAB EXERCISES

Date	@March 25, 2026

Name: Devadathan N R
Registration No: 230905010
Class: CSE - A1
Roll No: 04

Title: Form Processing using Django – Part II

Question 1

Create a Register page and Success page with the following requirements:

- Register page should contain four input TextBoxes for UserName, Password, Email id and Contact Number and also a button to submit. Make the username as compulsory field and other fields as optional.
- On button click, Success page is displayed with message "Welcome {UserName}" and also his Email and Contact Number has to be displayed.
- Use secure technique to send details to the Success page (Hint: use csrftoken) 4) Design a website with two pages.

Create project & App

```
# Create Django project
django-admin startproject web_project
cd web_project

# Create app
python manage.py startapp webapp
```

`web_project/settings.py`

```
import os # added at the top


INSTALLED_APPS = [
    'webapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `web_project/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('webapp.urls')),
]
```

App URLs: `webapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.register, name='register'),
    path('success/', views.success, name='success'),
]
```

View: `webapp/views.py`

```
from django.shortcuts import render

def register(request):
    return render(request, 'register.html')

def success(request):
    if request.method == 'POST':
        # Extract data from the POST request
        username = request.POST.get('username')
        email = request.POST.get('email')
        contact = request.POST.get('contact')

        context = {
            'username': username,
            'email': email,
            'contact': contact
        }
        return render(request, 'success.html', context)
    return render(request, 'register.html') # Redirect back if accessed directly
```

Create template/ in webapp/. Inside template/

`register.html`

```html
<!DOCTYPE html>
<html>
<head>
    <title>Register</title>
</head>
<body>
    <h2>Registration Page</h2>
    <form action="/success/" method="POST">
        {% csrf_token %}
        <label>User Name (Compulsory):</label><br>
        <input type="text" name="username" required><br><br>

        <label>Password (Optional):</label><br>
        <input type="password" name="password"><br><br>

        <label>Email ID (Optional):</label><br>
        <input type="email" name="email"><br><br>

        <label>Contact Number (Optional):</label><br>
        <input type="text" name="contact"><br><br>

        <button type="submit">Register</button>
    </form>
</body>
</html>
```

`success.html`

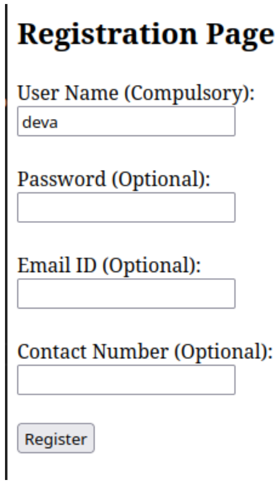
```html
<!DOCTYPE html>
<html>
<head>
    <title>Success</title>
</head>
<body>
    <h1>Welcome {{ username }}!</h1>
    <p><strong>Email:</strong> {{ email|default:"Not provided" }}</p>
    <p><strong>Contact:</strong> {{ contact|default:"Not provided" }}</p>
    <br>
    <a href="/">Go back to Register</a>
</body>
</html>
```

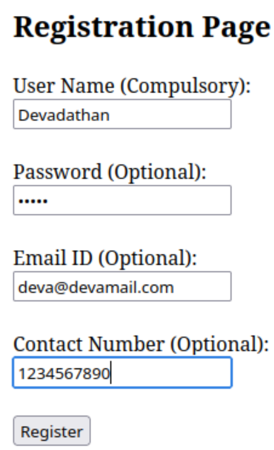
Run

```
python manage.py runserver 8080
```

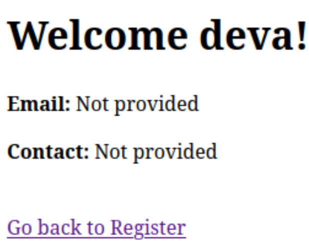
Output

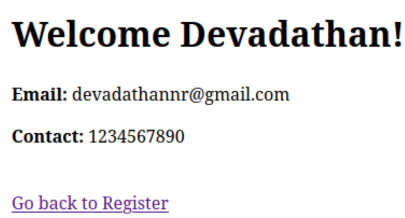
register.html





success.html





Question 2

“How is the book ASP.NET with c# by Vipul Prakashan?” Give the user three choice : i) Good ii) Satisfactory iii) Bad. Provide a VOTE button. After user votes, present the result in percentage using labels next to the choices

Create project & App

```
# Create Django project
django-admin startproject vote-project
cd vote-project

# Create app
python manage.py startapp voteapp
```

`vote_project/settings.py`

```
import os

INSTALLED_APPS = [
    'voteapp',
```

```
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `vote_project/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('voteapp.urls')),
]
```

App URLs: `voteapp/urls.py`

```
from django.urls import path
from . import views
from django.urls import path
from . import views

urlpatterns = [
    path('', views.firstPage, name='first'),
    path('submit/', views.secondPage, name='submit'),
    path('back/', views.goBack, name='back'),
]
urlpatterns = [
    path('', views.cover, name='cover'),
]
```

View: `voteapp/views.py`

```
from django.shortcuts import render

# Global dictionary to store vote counts (for lab simplicity)
# This data resets whenever the development server restarts.
vote_data = {
    'good': 0,
```

```
    'satisfactory': 0,
    'bad': 0,
    'total': 0,
}

def vote_page(request):
    context = {}
    if request.method == 'POST':
        # Get the value from the radio button group named 'choice'
        choice = request.POST.get('choice')

        # Verify that a valid choice was submitted
        if choice in ['good', 'satisfactory', 'bad']:
            # Increment the vote count
            vote_data[choice] += 1
            vote_data['total'] += 1

            # Calculate percentages (handle division by zero if total is 0)
            good_pct = (vote_data['good'] / vote_data['total'] * 100) if vote_data['total'] > 0
else 0
            satisfactory_pct = (vote_data['satisfactory'] / vote_data['total'] * 100) if vote_da
ta['total'] > 0 else 0
            bad_pct = (vote_data['bad'] / vote_data['total'] * 100) if vote_data['total'] > 0 el
se 0

            # Update context with results to display them on the page
            context = {
                'good_pct': round(good_pct, 1),
                'satisfactory_pct': round(satisfactory_pct, 1),
                'bad_pct': round(bad_pct, 1),
                'has_voted': True # Flag to potentially show results section
            }
        else:
             context = {'error_message': 'No choice selected'}

    return render(request, 'voting_page.html', context)
```

Create template/ in voteapp/. Inside template/

`voting_page.html`

```
<!DOCTYPE html>
<html>
<head>
    <title>Book Vote Lab</title>
</head>
<body style="background-color: #f0f2f5; font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-s
erif;">

    <center>
        <div style="background-color: white; width: 500px; margin-top: 50px; padding: 30px; bord
er-radius: 10px; border: 1px solid #ddd; box-shadow: 0px 4px 8px rgba(0,0,0,0.1); text-align: le
ft;">
```

```
            <h3 style="color: #333; margin-bottom: 25px; border-bottom: 2px solid #eee; padding-
bottom: 10px;">
                How is the book ASP.NET with c# by Vipul Prakashan?
            </h3>

            <form method="POST">
                {% csrf_token %}

                <table width="100%" cellpadding="10" cellspacing="0" border="0">
                    <tr>
                        <td width="40" valign="middle">
                            <input type="radio" name="choice" value="good" id="g" required style
="transform: scale(1.2); cursor: pointer;">
                        </td>
                        <td valign="middle">
                            <label for="g" style="font-size: 16px; cursor: pointer;">Good</label
>
                        </td>
                        <td align="right" valign="middle" style="color: #666; font-weight: bol
d;">
                            {{ good_pct|default:"0" }}%
                        </td>
                    </tr>

                    <tr>
                        <td valign="middle">
                            <input type="radio" name="choice" value="satisfactory" id="s" style
="transform: scale(1.2); cursor: pointer;">
                        </td>
                        <td valign="middle">
                            <label for="s" style="font-size: 16px; cursor: pointer;">Satisfactor
y</label>
                        </td>
                        <td align="right" valign="middle" style="color: #666; font-weight: bol
d;">
                            {{ satisfactory_pct|default:"0" }}%
                        </td>
                    </tr>

                    <tr>
                        <td valign="middle">
                            <input type="radio" name="choice" value="bad" id="b" style="transfor
m: scale(1.2); cursor: pointer;">
                        </td>
                        <td valign="middle">
                            <label for="b" style="font-size: 16px; cursor: pointer;">Bad</label>
                        </td>
                        <td align="right" valign="middle" style="color: #666; font-weight: bol
d;">
                            {{ bad_pct|default:"0" }}%
                        </td>
                    </tr>
                </table>
```

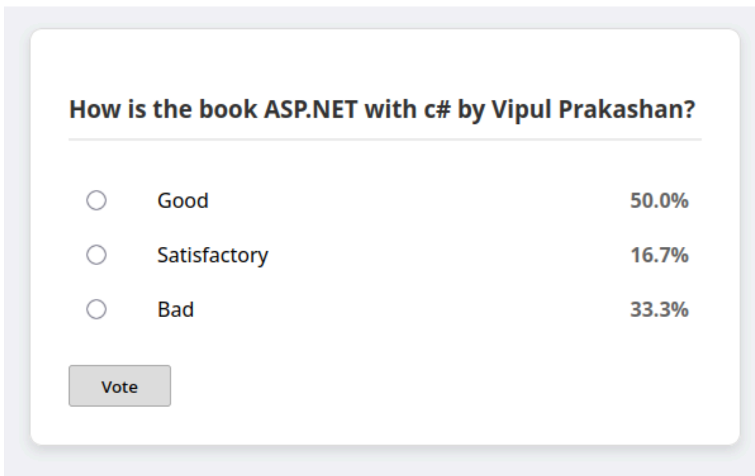
```
                 <br>
                 <div style="text-align: left;">
                     <input type="submit" value="Vote" style="background-color: #e0e0e0; border: 1px solid #adadad; padding: 6px 25px; cursor: pointer; border-radius: 3px; font-weight: 500;">
                 </div>
             </form>
         </div>
     </center>

</body>
</html>
```

Run

```
python manage.py runserver 8080
```

Output

voting_page.html



Question 3

Create a website with two pages. Page 1 has two TextBoxes (name and total marks) and one ‘Calculate’ Button as shown in the figure. On clicking the ‘Calculate’ Button, CGPA (total marks/50) along with the name should be displayed in the Page 2. Use Django sessions to store the information.

Create project & App

```
# Create Django project
django-admin startproject cgpa-project
cd cgpa-project

# Create app
python manage.py startapp cgpaapp
```

`cgpa-project/settings.py`

```
import os # added at the top


INSTALLED_APPS = [
    'cgpaapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `cgpa-project/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('cgpaapp.urls')),
]
```

App URLs: `cgpaapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.page1, name='page1'),
    path('result/', views.page2, name='page2'),
]
```

View: `cgpaapp/views.py`

```
from django.shortcuts import render, redirect

def page1(request):
    if request.method == 'POST':
        # Get data from form
```

```
        name = request.POST.get('name')
        marks = request.POST.get('marks')

        # Calculate CGPA
        cgpa = float(marks) / 50

        # Store in Sessions
        request.session['user_name'] = name
        request.session['user_cgpa'] = cgpa

        return redirect('page2')

    return render(request, 'page1.html')

def page2(request):
    # Retrieve data from Sessions
    name = request.session.get('user_name', 'Guest')
    cgpa = request.session.get('user_cgpa', 0)

    return render(request, 'page2.html', {'name': name, 'cgpa': cgpa})
```

Create template/ in cgpaapp/. Inside template/

`page1.html`

```
<!DOCTYPE html>
<html>
<body style="background-color: #f4f4f4; font-family: sans-serif;">
    <center>
        <div style="background: white; width: 400px; padding: 30px; margin-top: 50px; border: 1p
x solid #ccc; text-align: left;">
            <form method="POST">
                {% csrf_token %}
                <table border="0" cellpadding="10" width="100%">
                    <tr>
                        <td width="40%">Name:</td>
                        <td><input type="text" name="name" required style="width: 100%;"></td>
                    </tr>
                    <tr>
                        <td>Total Marks:</td>
                        <td><input type="number" name="marks" required style="width: 100%;"></td
>
                    </tr>
                </table>
                <br>
                <div style="text-align: center;">
                    <input type="submit" value="Calculate" style="padding: 5px 20px;">
                </div>
            </form>
            <p align="right" style="color: grey; font-size: 12px;">Page 1</p>
        </div>
    </center>
```

```
</body>
</html>
```

`page2.html`

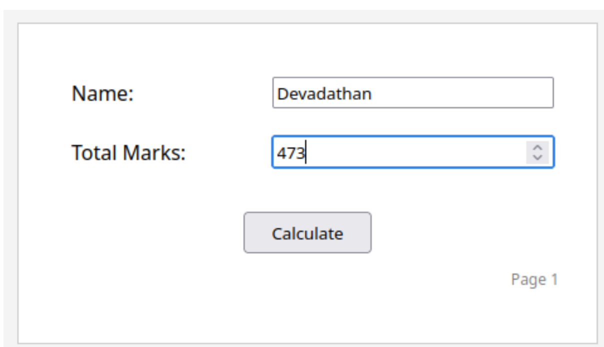
```
<!DOCTYPE html>
<html>
<body style="background-color: #f4f4f4; font-family: sans-serif;">
    <center>
        <div style="background: white; width: 400px; padding: 30px; margin-top: 50px; border: 1px solid #ccc;">
            <div style="text-align: left; padding: 20px;">
                <h2 style="margin: 0;">Welcome {{ name }}</h2>
                <h3 style="margin: 10px 0;">Your CGPA is = {{ cgpa }}</h3>
            </div>
            <br>
            <a href="{% url 'page1' %}">Go Back</a>
            <p align="right" style="color: grey; font-size: 12px;">Page 2</p>
        </div>
    </center>
</body>
</html>
```

Run

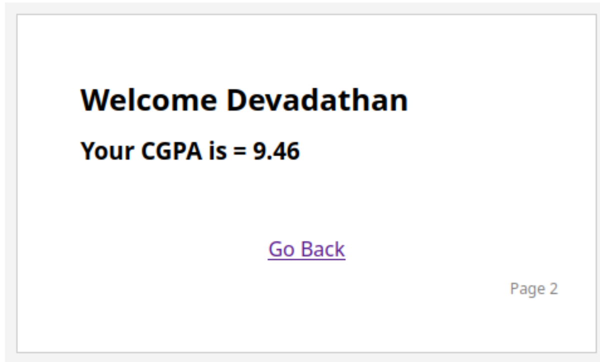
```
python manage.py runserver 8080
```

Output

page1.html



page2.html


Welcome Devadathan
Your CGPA is = 9.46
Go Back
Page 2